Recitation 1: Complex Numbers and Induction

Original recitation notes by Chen Gadi. English translation by Chen Gadi and ChatGPT.

These are recitations from previous years, provided for supplementary practice. Their numbering and topic order may differ from the current course. The material taught in the current lectures and official announcements is authoritative.

How to use these notes

These notes are intended for students in all fields of engineering. They combine definitions, guiding ideas, examples, and detailed solutions.

Goals of this recitation. We will learn to convert between Cartesian and polar forms, calculate powers and roots, and interpret operations in the complex plane. We will finish by writing a proof by induction, explicitly stating the range for which the claim holds.

A working habit. In every solution, state what is given, what is required, which definition or claim you are using, and why each operation is valid. Correct calculations and clear explanations complement one another.

1.1 Complex numbers: Cartesian form

Definition

The imaginary unit i satisfies $i^2 = -1$.
The expression $z = x + iy$ is a complex number, where $x, y \in \mathbb{R}$.

1. If $\mathrm{Re}(z) = x = 0$, then z is purely imaginary.
2. If $\mathrm{Im}(z) = y = 0$, then z is real.

Over $\mathbb{C}$, solve the equation $z^2 - 4z + 13 = 0$. Write a factorization into linear factors over $\mathbb{C}$, and determine whether such a factorization exists over $\mathbb{R}$.

Solution

The discriminant is $\Delta = (-4)^2 - 4 \cdot 1 \cdot 13 = 16 - 52 = -36$. Therefore,

$$z_{1,2} = \frac{4 \pm \sqrt{-36}}{2} = \frac{4 \pm 6i}{2} = 2 \pm 3i.$$

It follows that

$$z^2 - 4z + 13 = (z - (2 + 3i))(z - (2 - 3i)).$$

This is a factorization over $\mathbb{C}$. Over $\mathbb{R}$, the discriminant is negative, so there are no real roots and no factorization into two real linear factors.

Definition

- **Conjugate:** For $z = x + iy$, we write $z^* := x - iy$.
- **Modulus:** $|z| = \sqrt{x^2 + y^2} \in \mathbb{R}$.
- **Multiplicative inverse:** For $z \neq 0$: $z^{-1} = \dfrac{z^*}{|z|^2}$.

Prove: **a**. $\operatorname{Re} z = \dfrac{z + z^*}{2}$, $\operatorname{Im} z = \dfrac{z - z^*}{2i}$. **b**. $|z| \geq \operatorname{Im} z$, $|z| \geq \operatorname{Re} z$ for every $z \in \mathbb{C}$. **c**. (Homework) $(z^*)^* = z$. **d**. (Homework) If z is real, then $z^* = z$.

Solution
Write $z = x + iy$. Then

$$\frac{z + z^*}{2} = x = \operatorname{Re} z, \qquad \frac{z - z^*}{2i} = y = \operatorname{Im} z,$$

and

$$|z| = \sqrt{x^2 + y^2} \geq |x| \geq x = \operatorname{Re} z, \qquad |z| \geq |y| \geq y = \operatorname{Im} z.$$

(c) $(z^*)^* = (x - iy)^* = x + iy = z$.
(d) If z is real, then $y = 0$, so $z^* = x - 0i = x = z$.

1.1.1 Useful properties

1. $|z| = |z^*|$
2. $z \cdot z^* = |z|^2$ (Multiplication by the conjugate gives a real number)
3. $|zw| = |z|\,|w|$
4. $(zw)^* = z^* w^*$

Prove that, for $z \neq 0$: $(z^{-1})^* = (z^*)^{-1}$.

Solution
Since $z \neq 0$, we also have $z^* \neq 0$ and $|z|^2 > 0$. By the formula for the multiplicative inverse,

$$(z^{-1})^* = \left(\frac{z^*}{|z|^2}\right)^* = \frac{z}{|z|^2}, \qquad (z^*)^{-1} = \frac{(z^*)^*}{|z^*|^2} = \frac{z}{|z|^2}.$$

Both sides equal the same expression, so $(z^{-1})^* = (z^*)^{-1}$.

1.2 Polar form

1.2.1 Converting between Cartesian and polar forms

Definition

For every $z = x + iy \neq 0$, there exist $r = |z| > 0$ and an angle θ such that

$$z = r(\cos\theta + i\sin\theta) = re^{i\theta}, \qquad x = r\cos\theta, \quad y = r\sin\theta.$$

Conversely, $r = \sqrt{x^2 + y^2}$. When $x \neq 0$, we have $\tan\theta = y/x$, but this ratio alone does not determine the quadrant. **Remarks:**

- The angle is determined up to the addition of $2\pi k$, where $k \in \mathbb{Z}$. For $z = 0$, we have $r = 0$, and the argument is undefined.
- All angles in exponents $e^{i\theta}$ will be expressed in radians. Degrees will be used only for numerical illustration.

Convert $z = 3 + 3i$ and $w = -4 + 4i$ to polar form. Then write $a = 6e^{i\pi/3}$ in Cartesian form. In each polar form, choose an angle in the interval $[0, 2\pi)$.

Solution

For z, we have $|z| = \sqrt{3^2 + 3^2} = 3\sqrt{2}$. The components are equal and positive, so the angle is $\pi/4$:

$$z = 3\sqrt{2}e^{i\pi/4}.$$

For w, we have $|w| = \sqrt{(-4)^2 + 4^2} = 4\sqrt{2}$. The point lies in the second quadrant, and the ratio of the magnitudes of its components is 1, so

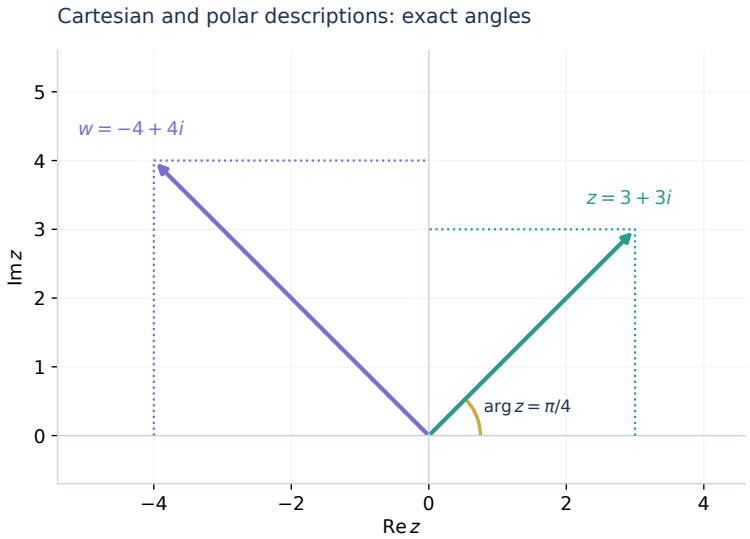
$$w = 4\sqrt{2}e^{3i\pi/4}.$$

Finally, we use Euler's identity and the trigonometric values at the angle $\pi/3$:

$$a = 6\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right) = 6\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) = 3 + 3\sqrt{3}i.$$

The angles are in radians. Both given numbers have positive modulus, so the issue of an argument at zero does not arise.

1.2.2 Arithmetic operations



Cartesian and polar forms in the same plane: the modulus is the distance from the origin, and the argument is the direction angle.

Definition

Multiplication: $(r_1e^{i\theta_1})(r_2e^{i\theta_2}) = r_1r_2\,e^{i(\theta_1+\theta_2)}$.

Division (when the denominator is $\neq 0$): $\dfrac{r_1e^{i\theta_1}}{r_2e^{i\theta_2}} = \dfrac{r_1}{r_2}\,e^{i(\theta_1-\theta_2)}$.

Conjugate: $(re^{i\theta})^* = r\,e^{-i\theta}$.

Periodicity: $re^{i(\theta+2\pi k)} = re^{i\theta}$, $re^{i(\theta+\pi)} = -re^{i\theta}$.

Let $z = 1 + i$ and $w = -4 + 4i$. Calculate zw and w/z in two ways: using polar form and using a Cartesian calculation. Check that the resulting moduli agree with the multiplication and division formulas.

Solution

Write $z = \sqrt{2}e^{i\pi/4}$ and $w = 4\sqrt{2}e^{3i\pi/4}$. Therefore,

$$zw = 8e^{i\pi} = -8, \qquad \frac{w}{z} = 4e^{i\pi/2} = 4i.$$

A Cartesian calculation gives

$$(1+i)(-4+4i) = -4 + 4i - 4i + 4i^2 = -8,$$

$$\frac{-4+4i}{1+i} = \frac{(-4+4i)(1-i)}{(1+i)(1-i)} = \frac{-4+8i+4}{2} = 4i.$$

Checking the moduli: $|z||w| = \sqrt{2}\cdot 4\sqrt{2} = 8 = |-8|$, and $|w|/|z| = 4 = |4i|$. The two calculation methods agree.

1.2.3 Raising to a power

Calculate $(1+i)^{12}$ using polar form. Verify the result using $(1+i)^4$.

Solution
Since $1+i=\sqrt{2}e^{i\pi/4}$,

$$(1+i)^{12}=(\sqrt{2})^{12}e^{12i\pi/4}=2^6e^{3i\pi}=-64.$$

To verify this, $(1+i)^2=2i$, so $(1+i)^4=(2i)^2=-4$. Hence $(1+i)^{12}=((1+i)^4)^3=(-4)^3=-64$.

1.2.4 Taking a complex root

Definition

Let $0\neq z\in\mathbb{C}$, and let $n\geq 1$ be an integer. The equation $w^n=z=re^{i\theta}$ has n distinct roots:

$$w_j=\sqrt[n]{r}\,e^{i\frac{\theta+2\pi j}{n}},\quad j=0,1,\ldots,n-1.$$

Geometric meaning: The roots lie on a circle of radius $\sqrt[n]{r}$, with equal angular spacing $2\pi/n$. When $n\geq 3$, they are the vertices of a regular polygon. For $n=1$ there is one root; for $n=2$ there are two opposite roots.

Find all solutions of $z^4=-i$.

Solution
Write $-i=e^{-i\pi/2}$. Therefore,

$$z_j=e^{i\frac{-\pi/2+2\pi j}{4}}=e^{i\frac{\pi(4j-1)}{8}},\quad j=0,1,2,3,$$

that is, the angles are $-\pi/8,\ 3\pi/8,\ 7\pi/8,\ 11\pi/8$ (at radius 1).

Let $n\geq 2$. Find all nth roots of unity and prove that their sum is zero. What happens when $n=1$?

Solution
The roots are $w_j=e^{i\frac{2\pi j}{n}}$, $j=0,\ldots,n-1$. Write $w=e^{i\frac{2\pi}{n}}$. Then $w_j=w^j$, so

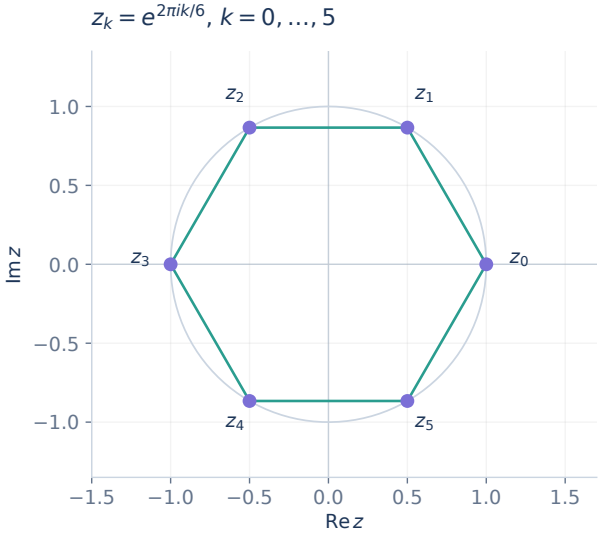
$$\sum_{j=0}^{n-1}w^j=\frac{w^n-1}{w-1}=0,$$

because $w^n=1$ and $w\neq 1$ when $n\geq 2$. For $n=1$, the only root is 1, so the sum is 1, not zero.

(**Homework**) Consider $a_nx^n+a_{n-1}x^{n-1}+\cdots+a_1x+a_0=0$ with real coefficients. Prove that if z is a solution, then z^* is also a solution.

Solution
We use the identities $(z_1 + \cdots + z_m)^* = z_1^* + \cdots + z_m^*$, $(z_1 \cdots z_m)^* = z_1^* \cdots z_m^*$, and the fact that a real number a satisfies $a^* = a$. If $a_0 + a_1 z + \cdots + a_n z^n = 0$, applying * gives $a_0 + a_1 z^* + \cdots + a_n (z^*)^n = 0$. Thus z^* is a solution.

1.3 Further properties of complex numbers



Roots of unity are equally spaced in angle on the unit circle; for an order greater than 1, their sum is zero.

1.3.1 Multiplication by i

Multiplication by i rotates a complex number counterclockwise through an angle of $\frac{\pi}{2}$.

Prove that multiplication by i produces a rotation through an angle of $\frac{\pi}{2}$.

Solution
Cartesian representation:
Suppose $z = a + bi$. Multiplication by i gives

$$z \cdot i = (a + bi)i = -b + ai.$$

The vector (a, b) has been replaced by the vector $(-b, a)$. This is exactly a counterclockwise rotation through an angle of $\pi/2$ about the origin: the new x coordinate is $-b$, and the new y coordinate is a.
Polar representation:
The case $z = 0$ is immediate: the origin remains fixed. For a nonzero complex number, suppose $z = re^{i\theta}$, where $r = |z|$ and θ is the angle that z makes with the positive x axis. Multiplication by i can be written as

$$z \cdot i = re^{i\theta} \cdot i = re^{i\theta} \cdot e^{i\frac{\pi}{2}} = re^{i(\theta + \frac{\pi}{2})}.$$

Multiplication by i increases θ by the angle $\frac{\pi}{2}$, meaning that the complex number z rotates through an angle of $\frac{\pi}{2}$ about the origin, as required.

1.3.2 Division in polar form

Definition

If $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$, and also $r_2 \neq 0$, then

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}.$$

Calculate $\frac{z_1}{z_2}$ for $z_1 = 4e^{i\frac{\pi}{3}}$ and $z_2 = 2e^{i\frac{\pi}{6}}$.

Solution
We calculate:

$$\frac{z_1}{z_2} = \frac{4}{2} e^{i\left(\frac{\pi}{3} - \frac{\pi}{6}\right)} = 2e^{i\frac{\pi}{6}}.$$

Let $z_1, z_2 \neq 0$. Prove that an argument of the quotient z_1/z_2 equals the difference of the arguments, up to a multiple of 2π.

Solution
Suppose $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$, where $r_1, r_2 > 0$. Division gives

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}.$$

Hence $\theta_1 - \theta_2$ is an argument of the quotient. If an argument in a fixed range is required, add a suitable multiple of 2π.

1.3.3 Choosing an argument in a fixed range

In these notes, $\mathrm{Arg}(z)$ denotes the unique argument in the range $[0, 2\pi)$, for $z \neq 0$.

Remark

We cannot always write $\mathrm{Arg}(z_1 \cdot z_2) = \mathrm{Arg}(z_1) + \mathrm{Arg}(z_2)$ or $\mathrm{Arg}\left(\frac{z_1}{z_2}\right) = \mathrm{Arg}(z_1) - \mathrm{Arg}(z_2)$, because the result may fall outside the range $[0, 2\pi)$.

Calculate $\mathrm{Arg}(z_1 \cdot z_2)$ for $z_1 = e^{i\frac{3\pi}{4}}, z_2 = e^{i\frac{5\pi}{4}}$.

Solution
First, calculate the product itself:

$$z_1 z_2 = e^{i3\pi/4} e^{i5\pi/4} = e^{i2\pi} = 1.$$

The angle 2π is an argument of 1, but it is not in the range $[0, 2\pi)$. Therefore,

$$\mathrm{Arg}(z_1 z_2) = \mathrm{Arg}(1) = 0, \qquad \mathrm{Arg}(z_1) + \mathrm{Arg}(z_2) = 2\pi.$$

The two numbers are not equal; they are congruent modulo 2π.

1.3.4 Calculating the argument and analyzing quadrants

Definition

For a complex number $z = a + bi \neq 0$, where $a \neq 0$, every argument θ satisfies

$$\tan(\theta) = \frac{b}{a}.$$

To determine θ precisely, we must take into account the quadrant in which the complex number lies:

- If $a > 0, b > 0$: the angle is $\theta = \arctan\left(\frac{b}{a}\right)$.
- If $a < 0, b > 0$: add π to account for the second quadrant: $\theta = \pi + \arctan\left(\frac{b}{a}\right)$.
- If $a < 0, b < 0$: add π for the third quadrant: $\theta = \pi + \arctan\left(\frac{b}{a}\right)$.
- If $a > 0, b < 0$: add 2π to obtain a positive angle in the fourth quadrant: $\theta = 2\pi + \arctan\left(\frac{b}{a}\right)$.

If $b = 0$ and $a > 0$, then $\mathrm{Arg}(z) = 0$; if $b = 0$ and $a < 0$, then $\mathrm{Arg}(z) = \pi$. If $a = 0$, the angle depends only on the value of b:

- If $b > 0$: $\theta = \frac{\pi}{2}$.
- If $b < 0$: $\theta = \frac{3\pi}{2}$.

Find the angle θ for $z = -1 + \sqrt{3}i$.

Solution

We are given $a = -1, b = \sqrt{3}$. First, calculate

$$\tan(\theta) = \frac{\sqrt{3}}{-1} = -\sqrt{3}.$$

Since $a < 0$ and $b > 0$, the number lies in the second quadrant. Therefore,

$$\theta = \pi - \frac{\pi}{3} = \frac{2\pi}{3}.$$

Find the angle θ for $z = -5 - 5i$.

Solution

We are given $a = -5, b = -5$. First, calculate

$$\tan(\theta) = \frac{b}{a} = \frac{-5}{-5} = 1.$$

Since $a < 0, b < 0$, the number lies in the third quadrant. Therefore,

$$\theta = \pi + \arctan(1) = \pi + \frac{\pi}{4} = \frac{5\pi}{4}.$$

1.4 The principle of induction

Definition

The principle of (weak) induction:
Let n_0 be a nonnegative integer, and let $P(n)$ be a statement defined for every integer $n \geq n_0$. Suppose that

1. **Base case:** The statement $P(n_0)$ is true.
2. **Induction step:** For every integer $n \geq n_0$, the assumption $P(n)$ implies the statement $P(n+1)$.

Then $P(n)$ is true for every integer $n \geq n_0$.

Prove by induction that, for every $n \geq 1$,

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Solution
Defining the statement: $P(n): \sum_{i=1}^{n} i = \frac{n(n+1)}{2}$.
Base case: $n = 1$: the left-hand side $= 1$, and the right-hand side $= \frac{1 \cdot 2}{2} = 1$. The statement is true.
Induction hypothesis: Suppose that, for some $k \geq 1$, we have $\sum_{i=1}^{k} i = \frac{k(k+1)}{2}$.
Induction step:

$$\sum_{i=1}^{k+1} i = \left(\sum_{i=1}^{k} i\right) + (k+1) = \frac{k(k+1)}{2} + (k+1) = \frac{(k+1)(k+2)}{2}.$$

We have obtained the formula for $k+1$.
Conclusion: The statement is true for every $n \geq 1$.

Prove that, for every $n \geq 1$, we have $7 \mid (8^n - 1)$.

Solution
Defining the statement: $P(n): 7 \mid (8^n - 1)$.
Base case: $n = 1$: $8^1 - 1 = 7$, and indeed $7 \mid 7$.
Induction hypothesis: Suppose that, for some $k \geq 1$, we have $7 \mid (8^k - 1)$; that is, there exists $m \in \mathbb{Z}$ such that $8^k - 1 = 7m$.
Induction step:

$$8^{k+1} - 1 = 8 \cdot 8^k - 1 = 8(8^k - 1) + 7.$$

By the hypothesis, $8^k - 1 = 7m$, so

$$8^{k+1} - 1 = 8 \cdot 7m + 7 = 7(8m + 1),$$

and therefore $7 \mid (8^{k+1} - 1)$.
Conclusion: For every $n \geq 1$, we have $7 \mid (8^n - 1)$.
A shorter argument: $8 \equiv 1 \pmod 7 \Rightarrow 8^n \equiv 1 \pmod 7$.

Prove that, for every $n \geq 4$,

$$2^n \geq n^2.$$

Solution
We prove this by induction on n.
Base case: $n = 4$: $2^4 = 16 = 4^2$. The statement is true.
Induction hypothesis: Suppose $2^k \geq k^2$ for $k \geq 4$.
Induction step:

$$2^{k+1} = 2 \cdot 2^k \;\geq\; 2k^2.$$

It suffices to show that $2k^2 \geq (k+1)^2 \iff k^2 - 2k - 1 \geq 0$. For $k \geq 4$, we have $k^2 - 2k - 1 > 0$, so $2^{k+1} \geq (k+1)^2$.
Hence, by induction, $2^n \geq n^2$ for every $n \geq 4$.

1.5 Supplementary exercises: geometry and signal calibration

Find and describe geometrically all $z \in \mathbb{C}$ satisfying $|z - 1| = |z + 1|$. Which of them also satisfy $|z| = 2$?

Solution
Write $z = x + iy$. Both sides are nonnegative, so squaring is valid:

$$|z-1|^2 = (x-1)^2 + y^2, \qquad |z+1|^2 = (x+1)^2 + y^2.$$

It follows that

$$(x-1)^2 + y^2 = (x+1)^2 + y^2 \iff x^2 - 2x + 1 = x^2 + 2x + 1 \iff x = 0.$$

This is the imaginary axis: the set of points equidistant from the two points $1, -1$. The additional condition gives $y^2 = 4$, so the intersection consists of $z = 2i$ or $z = -2i$.

In a simple model of a signal with two components, a calibration system performs the operation $T(z) = (1 + i)z$. What is the effect on the magnitude and angle? Calculate $T(1 - i)$, and find an input for which the output is $2i$.

Solution
Since $1 + i = \sqrt{2}e^{i\pi/4}$, for every $z = re^{i\theta} \neq 0$, we have $T(z) = \sqrt{2}\, r\, e^{i(\theta+\pi/4)}$. Thus, the magnitude is multiplied by $\sqrt{2}$, and the direction rotates counterclockwise through an angle of $\pi/4$. We also have $T(0) = 0$. A direct calculation gives $T(1 - i) = (1 + i)(1 - i) = 1 - i^2 = 2$. To obtain $2i$, solve $(1 + i)z = 2i$:

$$z = \frac{2i}{1+i} = \frac{2i(1-i)}{(1+i)(1-i)} = \frac{2i+2}{2} = 1 + i.$$

Check: $(1 + i)^2 = 1 + 2i + i^2 = 2i$.